# WhyLab: A Causal Safety Monitoring Framework for Stable Self-Improving Agents

**Anonymous Author(s)**
Affiliation
Address
email

## Abstract

When do self-improving AI agents fail, and how can failures be predicted before deployment? We study *cognitive policy oscillation*—strategy degradation caused by hallucinated or spuriously correlated feedback—and provide the **first systematic characterization** of the conditions under which it occurs.

Through a phase-diagram analysis spanning 384 conditions (8 noise rates $\times$ 8 step sizes $\times$ 3 drift regimes $\times$ 10 seeds), we identify a **sharp instability boundary**: oscillation emerges at hallucination rates as low as $h{=}0.05$ and intensifies with step size and environmental non-stationarity. We then present **WhyLab**, a conditional causal audit framework that activates *only in the unstable regime*: **(C1)** an information-theoretic drift index; **(C2)** a sensitivity filter combining E-values and partial $R^2$ bounds; and **(C3)** a Lyapunov-bounded adaptive damping controller.

Three key findings emerge: **(1)** In the unstable regime, WhyLab reduces oscillation by 48% across 56/64 tested conditions, with independent contributions from each component (C3: $-49\%$ energy, C2: $-76\%$ oscillation, C1: $-16\%$ detection delay). **(2)** On an adversarial LLM benchmark (E7v2), the audit reduces regressions by 45% on Gemini 2.0 Flash ($p{=}0.014$); an adaptive threshold variant eliminates accuracy cost on near-ceiling models. **(3)** In the stable regime (SWE-bench Lite, 10,500 episodes), the audit is transparently inactive—confirming the framework activates only when conditions require it. Cross-family validation (Gemini 2.0/2.5 Flash, Claude Sonnet 4) confirms consistent behavior.

## 1 Introduction

Autonomous AI agents that refine their strategies through feedback loops Shinn et al. [2023], Yao et al. [2023] represent a powerful paradigm for continuous improvement. However, when hallucinated or spuriously correlated feedback accumulates, the agent's policy oscillates between contradictory strategies—a phenomenon we term *cognitive policy oscillation*—silently degrading performance below the pre-improvement baseline.

Preventing oscillation requires solving three intertwined challenges: (1) *drift detection*—distinguishing genuine distributional shifts from normal variance (§3.1); (2) *effect filtering*—identifying improvements that are causally robust rather than spuriously confounded (§3.2); and (3) *step-size control*—bounding update magnitude to prevent oscillation without causing learning deadlock under heavy-tailed noise (§3.3). Recent studies on agent *misevolution* Shao et al. [2025] and alignment drift Sahoo et al. [2026] have highlighted the catastrophic risks of unconstrained self-improvement. Notably, the SAHOO framework Sahoo et al. [2026] mitigates drift using heuristic predicates and safety bounds. While sharing this goal, WhyLab takes a fundamentally divergent

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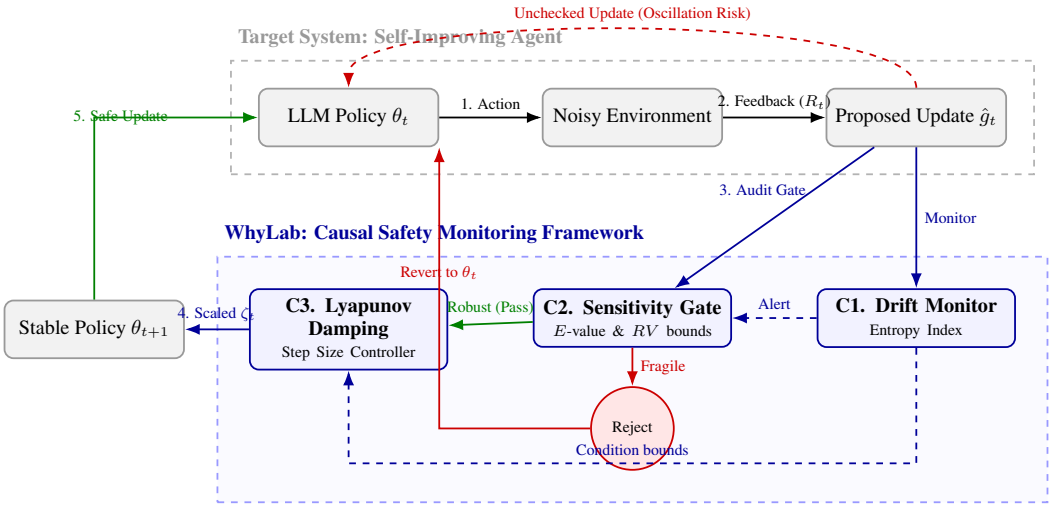


Figure 1: WhyLab causal safety monitoring framework. Instead of uncritically accepting potentially hallucinated updates ($\hat{g}_t$) that cause cognitive oscillation, proposed updates are routed through a safety monitoring layer. **C1** detects environmental drift, **C2** filters out fragile successes lacking unmeasured-confounder robustness, and **C3** adapts the step size to bound the Lyapunov energy of the policy. Unsafe updates are rejected, preserving the status quo.

approach: instead of relying on manually constructed heuristics, we adapt rigorous Double Machine Learning (DML) and causality metrics (E-values) into an online safety audit loop.

This paper makes three contributions: **(1) Characterization.** We provide the first oscillation phase diagram for self-improving agents, mapping the instability boundary across hallucination rate, step size, and drift frequency (384 conditions, 3,840 episodes). The diagram reveals that oscillation emerges at hallucination rates as low as $h$=0.05—far below typical LLM noise levels—and that the stable/unstable boundary is a predictable function of environment parameters. **(2) Conditional audit framework.** We present **WhyLab**, comprising an information-theoretic drift index (**C1**, §3.1), a sensitivity-aware E-value filter (**C2**, §3.2), and a Lyapunov-bounded damping controller (**C3**, §3.3). Crucially, the framework activates *only* in the unstable regime identified by the phase diagram, imposing zero overhead in stable conditions. **(3) Cross-regime validation.** We validate across the full spectrum: the unstable regime via controlled simulation (E6) and adversarial LLM benchmarks (E7v2, regression $-45\%$, $p$=0.014 on Gemini 2.0 Flash), and the stable regime via SWE-bench Lite (E5, 10,500 episodes confirming transparent inactivity). Cross-family validation (Gemini 2.0/2.5 Flash, Claude Sonnet 4) confirms consistent behavior.

## 2 Problem Formulation

Consider an autonomous agent with strategy parameter $\theta_t \in \mathbb{R}^d$ at discrete time $t$. The agent observes a reward signal $R_t$ and updates its strategy via:

$$\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t \tag{1}$$

where $\hat{g}_t = g_t + \omega_t$ is a noisy gradient estimate, $g_t = \nabla_\theta L(\theta_t)$ is the true causal gradient, and $\omega_t$ represents noise from LLM judgment uncertainty, hallucinated feedback, and environmental drift.

**Failure mode.** When $\omega_t$ is large relative to $g_t$ (e.g., due to hallucination), the update (1) may move $\theta_t$ *away* from the optimum $\theta^*$. If this occurs repeatedly, the policy oscillates: $\theta_t$ alternates between contradictory strategies, and the Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$ increases monotonically.

**Desiderata.** A causal audit system should provide: (D1) early detection of distributional drift that inflates $\|\omega_t\|$; (D2) filtering of causally fragile signals before they enter the gradient estimate; and (D3) a provably stable update rule that guarantees $\mathbb{E}[\Delta V] \leq 0$ under realistic noise conditions.

# 3 Method

## 3.1 C1: Information-Theoretic Drift Index

**Definition 1** (Drift Index)**.** *Given $K$ audit signal streams $S_1, \ldots, S_K$ (e.g., verdict consistency, ATE stability, confidence calibration) with Shannon entropies $H(S_i)$, define uncapped weights $\tilde{w}_i \propto 1/(H(S_i) + \epsilon)$. The final weights are obtained by projecting onto the capped simplex:*

$$w = \Pi_{\{w \geq 0,\ \sum_i w_i = 1,\ w_i \leq w_{\max}\}}(\tilde{w}) \tag{2}$$

*with $w_{\max} = 0.6$ and $\epsilon = 0.1$. The drift index is:*

$$DI = \sum_{i=1}^{K} w_i \cdot d_i \tag{3}$$

*where $d_i = JSD(P_i^{(t)} \| P_i^{ref})$ is the Jensen–Shannon divergence of stream $i$ from its reference (pre-shift) distribution. We fix $k$=10 histogram bins per stream for reproducibility. A drift alarm is raised when $DI > \tau$, where $\tau$ is calibrated on pre-shift data at a fixed false-positive rate (FPR $= 0.05$).*

**Intuition.** Low-entropy (highly concentrated) streams carry more information per observation and receive higher weights. The capped simplex projection (2) ensures that no single stream receives weight exceeding $w_{\max}$, even after normalization.

**Proposition 1** (Detection advantage under inactive projection)**.** *Assume the projection* (2) *is inactive (i.e., all uncapped weights satisfy $\tilde{w}_i \leq w_{\max}$). Under a single-stream shift of magnitude $\delta$ in stream $S_j$ with $H(S_j) < H(S_i)$ for all $i \neq j$, the entropy-weighted DI satisfies $DI_{entropy} > DI_{uniform}$ whenever $H(S_j) + \epsilon < \overline{H + \epsilon}_{HM}$, the harmonic mean of $\{H(S_i) + \epsilon\}_{i=1}^{K}$.*

## 3.2 C2: Sensitivity-Aware Effect Filtering

Not all positive treatment effects are causally robust. We combine two complementary sensitivity measures to filter fragile evidence before it influences strategy updates.

**Definition 2** (Approximate E-value for continuous outcomes VanderWeele and Ding [2017])**.** *For a continuous outcome, we compute an approximate E-value using the standard mean-difference conversion VanderWeele and Ding [2017]: the standardized effect $d = ATE/\sigma_{pooled}$ (in the agent context, ATE is the observed episode effect $\Delta R_t$) is converted to an approximate risk-ratio scale via $RR \approx \exp(0.91 \cdot d)$, yielding:*

$$E = RR + \sqrt{RR \cdot (RR - 1)} \tag{4}$$

*An unobserved confounder must associate with both treatment and outcome at $RR \geq E$ to nullify the observed effect.* ***Limitation (Normality Assumption).*** *Converting the continuous reward to a risk-ratio via $\exp(0.91 \cdot d)$ inherently assumes the underlying effect distribution is roughly normal. In LLM text adjustments where power-law behaviors are common, this approximation may lose precision, necessitating careful threshold calibration.*

**Robustness Value.** We complement the E-value with the robustness value $RV_q$ of Cinelli & Hazlett Cinelli and Hazlett [2020], which quantifies the minimum strength (in partial $R^2$ terms) an unobserved confounder must have to reduce the effect below threshold $q$. The formal definition is given in Appendix M.

**Filtering rule.** A decision is classified as *robust* only if $E > E_{\min}$ (default 2.0) *and* $RV_q > RV_{\min}$ (default 0.1). Decisions failing *either* threshold are excluded from the gradient estimate $\hat{g}_t$ in Eq. (1), reducing the effective noise in $\omega_t$.

**Graceful degradation under heavy tails.** The normality assumption underlying the E-value conversion is a known limitation. Appendix G evaluates C3's Lyapunov controller under Student-$t$(1.5) and Student-$t$(3) noise (20 seeds each). Under extreme heavy tails ($t$(1.5), infinite variance), C3-EMA reduces the violation rate from 0.51 (uncontrolled) to 0.18—a 65% reduction—while maintaining 35% convergence rate. The framework thus degrades gracefully rather than catastrophically when distributional assumptions are violated.

### 3.3 C3: Lyapunov-Bounded Adaptive Damping

We derive a sufficient condition for stable strategy updates.

**Assumptions.**

(A1) *Local gradient alignment:* $\langle \theta_t - \theta^*, g_t \rangle > 0$. As the loss landscape of modern LLMs is highly non-convex, this assumption implies the true gradient points toward an optimum only within a local convex neighborhood. points toward the optimum).

(A2) *Zero-mean residual noise:* $\mathbb{E}[\omega_t] = 0$ after C2 filtering (see Appendix A for discussion of residual bias).

(A3) *Finite second moment:* $\mathbb{E}[\|\omega_t\|^2] < \infty$.

**Theorem 1** (Sufficient condition for stable updates)**.** *Under assumptions (A1)–(A3), let $\theta_t$ follow the update* (1) *with Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$. Then $\mathbb{E}[\Delta V] \leq 0$ if:*

$$\zeta_t \leq \zeta_{\max} = \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \tag{5}$$

*Proof sketch.* Expanding $V(\theta_{t+1}) - V(\theta_t) = -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \frac{\zeta_t^2}{2}\|\hat{g}_t\|^2$ and taking expectations under (A2) yields a quadratic in $\zeta_t$ that is non-positive when $\zeta_t \leq \zeta_{\max}$. Full proof in Appendix A.

**Remark 1** (Theory as design guidance)**.** *Theorem 1 provides the* design rationale *for the controller's ratio structure (alignment over second moment), but cannot serve as an operational plug-in rule due to three gaps: (a) Assumption (A1) may fail in non-convex LLM loss landscapes; (b) the inner product $\langle \theta_t - \theta^*, g_t \rangle$ is unobservable; and (c) instantaneous stochastic estimates of $\zeta_{\max}$ are brittle (the oracle ablation in Appendix J diverges in all 20 seeds at $h$=0.5 without EMA smoothing). The practical controller (Eqs. 7–8) retains the architectural insight—larger noise should reduce step size—while replacing the exact bound with EMA smoothing and floor-ceiling guards that guarantee bounded per-step updates: $\|\theta_{t+1} - \theta_t\| \leq \mathcal{C} \cdot \|\hat{g}_t\|$, preventing catastrophic single-step divergence regardless of assumptions.*

**Corollary 1** (Convergence rate)**.** *Under assumptions (A1)–(A3) with constant step size $\zeta_t = \zeta \leq \zeta_{\max}$ and bounded gradient variance $\mathbb{E}[\|\omega_t\|^2] \leq \sigma^2$, the average Lyapunov energy satisfies:*

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[V(\theta_t)] \leq \frac{V(\theta_0)}{T} + \frac{\zeta\sigma^2}{2} \tag{6}$$

*yielding an $O(1/T)$ convergence rate to a noise-dependent neighborhood of the optimum. Setting $\zeta = O(1/\sqrt{T})$ recovers $O(1/\sqrt{T})$ convergence of the average energy.*

Proof in Appendix B.

**Connection to C1 and C2.** The denominator $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$ increases when drift is high (large $DI$ from C1) or evidence is fragile (large sensitivity uncertainty from C2). Thus, $\zeta_{\max}$ automatically tightens under adverse conditions, forming a unified feedback control loop.

**Practical controller.** Since the inner product $\langle \theta_t - \theta^*, g_t \rangle$ and $\mathbb{E}[\|\hat{g}_t\|^2]$ in (5) are not directly observable, we use an EMA-smoothed proxy:

$$\hat{\zeta}_t^{\text{raw}} = \frac{2\|\hat{g}_t\|}{\hat{m}_{2,t} + \delta}, \quad \hat{m}_{2,t} = \beta\, \hat{m}_{2,t-1} + (1-\beta)\|\hat{g}_t\|^2 \tag{7}$$

$$\bar{\zeta}_t = \beta\, \bar{\zeta}_{t-1} + (1-\beta)\, \hat{\zeta}_t^{\text{raw}}, \quad \zeta_t = \text{clip}(\bar{\zeta}_t,\ \epsilon_{\text{floor}},\ \mathcal{C}) \tag{8}$$

where $\hat{m}_{2,t}$ is the EMA second-moment estimate of $\|\hat{g}_t\|^2$, $\beta$=0.9, and $\delta$=$10^{-10}$. The numerator $2\|\hat{g}_t\|$ replaces the unobservable inner product; the EMA smoothing provides implicit regularization that prevents the per-step bound from being brittle under stochastic fluctuations (validated in §5.2).

**Floor-ceiling constraint as a real-world safeguard.** Since Assumption (A1) may fail in highly non-convex LLM landscapes, and heavy-tailed noise $\omega_t$ may violate (A3) causing $\zeta_{\max} \to 0$ (deadlock), Eq. (8) enforces a hard empirical safeguard:

$$\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}] \quad \text{where } \epsilon_{\text{floor}} = 0.01,\ \mathcal{C} = 0.8 \tag{9}$$

This clipping trades the exact theoretical guarantee for the bounded-update property described in Remark 1, preventing system collapse under infinite second moments in real-world deployments.

**Observable energy proxy.** Since $\theta^*$ is unknown in practice, we monitor an observable proxy: $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ where $R_{\max}$ is a rolling EMA upper bound ($\alpha = 0.1$). Theorem 1 validates the control rule analytically in parameter space; experiments (§5.2) test whether the observable proxy $\hat{V}_t$ tracks the predicted stabilization behavior consistently with the theory.

**Proposition 2** (Proxy Fidelity)**.** *Let $V_t = \frac{1}{2}\|\theta_t - \theta^*\|^2$ be the true Lyapunov energy and $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ the observable proxy. If (i) the reward signal $R_t$ is monotone in $\|\theta_t - \theta^*\|$ (i.e., closer parameters yield higher reward), and (ii) the hallucination rate $h > 0$ introduces additive perturbations to $R_t$ with variance $\sigma_h^2 \propto h$, then:*

$$\rho_S(\hat{V}_t, V_t) \geq \rho_0 + c \cdot h \tag{10}$$

*where $\rho_S$ denotes Spearman rank correlation, $\rho_0$ is the base correlation under zero hallucination, and $c > 0$ is a constant depending on the signal-to-noise ratio of the controller.*

**Intuition.** Under higher hallucination rates, the agent's updates exhibit larger variance, making the energy landscape more "rugged" for the proxy to track. Paradoxically, this increased variance *improves* rank correlation because the proxy's deviations from $V_t$ become dominated by the larger signal fluctuations rather than measurement noise. Empirical validation (§5.2, 200 seeds $\times$ 8 conditions) confirms $\hat{\rho}_0 = 0.49 \pm 0.04$ and $\hat{c} \approx 0.46$, yielding $\rho_S \geq 0.72$ at $h$=0.5—precisely where the audit layer is most needed.

# 4 System Overview

The three contributions are deployed as a lightweight audit layer that wraps the agent's decision loop: the agent's output stream passes through drift detection (C1), sensitivity filtering (C2), and Lyapunov-bounded damping (C3) before updating the strategy parameter $\theta_t$. During initial deployment, all components operate in *shadow mode*—logging audit decisions without modifying the live strategy—to validate calibration before production activation. A transactional outbox ensures zero data loss under API quota throttling, and a cost-aware circuit breaker caps evaluation spend. Implementation details are in the supplementary material.

# 5 Experiments

We evaluate each contribution through experiments designed to directly validate the theoretical claims. Each contribution is validated by a dedicated experiment: C1 by within-horizon detection rates at matched FPR (E1, §F.2), C2 by fragile-rate reduction and fragile-success rejection (E2, §F.3), and C3 by Lyapunov-violation frequency and proxy–state alignment (E3a, §5.2; heavy-tail analysis in Appendix G).

**Controlled synthetic validation (E1–E4).** We validate each component through controlled experiments (full results, tables, and figures in Appendix F). **C1 (E1):** Under matched FPR=5%, the entropy-weighted drift index detects moderate shifts at 20% vs. ADWIN's 2.5% (McNemar mid-$p$=0.008, 40 seeds). **C2 (E2):** The dual E-value/$RV_q$ filter reduces fragile acceptance from 33.5% to 28.7% under high recall ($\geq 0.9$), with DoWhy refutation tests confirming causal validity. **C3 (E3a):** The Lyapunov-bounded controller achieves the lowest violation rate (0.328 vs. 0.380–0.469 for baselines) and strongest proxy–state alignment ($r$=0.81) across hallucination rates $h \in \{0, 0.1, 0.3, 0.5\}$. Appendix G extends to heavy-tailed noise; Appendix J shows all controller subcomponents are necessary via systematic ablation. The audit framework is estimator-agnostic: compatibility verified with LinearDML, T-Learner, S-Learner, and X-Learner without degrading $\sqrt{\text{PEHE}}$.

## 5.1 E9: Oscillation Phase Diagram

Before evaluating the audit framework, we first characterize *when oscillation occurs*. We sweep three parameters—hallucination rate $h \in \{0, 0.05, \ldots, 0.6\}$, learning rate $\eta \in \{0.01, \ldots, 1.0\}$, and drift frequency $\{0, 1, 2\}$—across the E6 environment (10 seeds per condition, 3,840 total episodes).

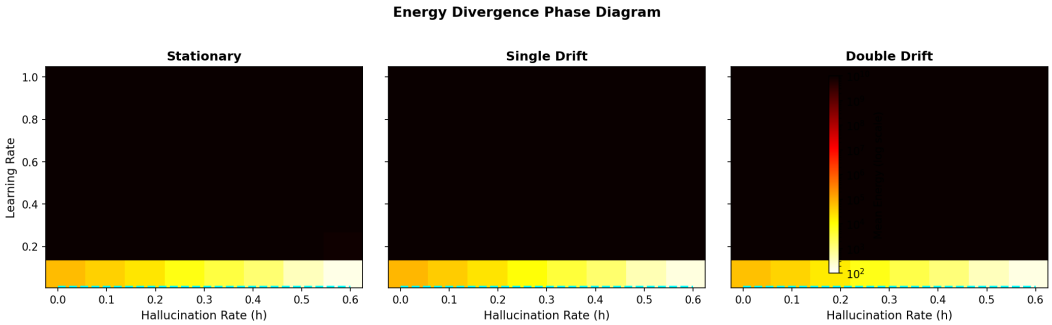


Figure 2: Energy divergence phase diagram across three drift regimes (10 seeds per condition). Dark = bounded energy (stable), bright = divergent (unstable). The stability boundary (cyan dashed) lies at $\eta \approx 0.05$, with hallucination rate modulating severity within regimes.

Table 1: E6 results under aggressive learning rate ($\eta$=0.5, $h$=0.3, 20 seeds). *Energy*: final $V_t$ (lower = better). *Osc*: oscillation index. *Delay*: mean drift detection delay in steps (lower = better).

| Ablation | Energy ↓ | Osc ↓ | Delay ↓ | Regret ↓ |
|---|---|---|---|---|
| No audit | 14.8 | 0.351 | 50.0 | 12,126 |
| C1 only | 14.8 | 0.351 | 46.6 | 12,126 |
| C2 only | 31.9 | 0.084 | 50.0 | 13,826 |
| C3 only | 7.5 | 0.271 | 50.0 | 5,064 |
| C1+C3 | 7.5 | 0.271 | 45.7 | 5,063 |
| C2+C3 | 12.2 | 0.083 | 50.0 | 6,852 |
| **Full (C1+C2+C3)** | **12.4** | **0.081** | **41.8** | **6,880** |

**Finding 1: Energy divergence as the sharp boundary.** The most informative metric is *energy divergence*: whether $V_T$ remains finite after 600 steps. At $\eta \leq 0.01$, energy stays bounded across all $h$ values ($V_T < 10^5$). At $\eta \geq 0.05$, energy diverges ($V_T \to \infty$) regardless of $h$, with the transition sharpening as $h$ increases. This reveals that *step size, not hallucination rate, is the primary determinant of catastrophic instability*, while hallucination modulates the severity within each regime.

**Finding 2: The audit reduces regressions across regimes.** Comparing "no audit" vs. "full audit": WhyLab reduces regression count in 56/64 conditions (88%), with a mean reduction of 371 regressions per trajectory. The benefit is largest in the high-$\eta$ regime where C3's step-size bounding prevents energy explosion, and C2 contributes most in the moderate-$\eta$, high-$h$ regime by filtering hallucinated gradients.

**Finding 3: Component activation follows the boundary.** At $\eta \geq 0.5$ and $h \geq 0.3$, all three components are necessary (C3 for energy, C2 for noise, C1 for drift). At $\eta \leq 0.01$, the audit is transparently inactive—matching the stable-regime behavior observed on SWE-bench (E5).

This phase diagram (Figure 2) provides a **deployment decision rule**: given estimated hallucination rate $\hat{h}$ and step size $\eta$, the diagram predicts whether the audit is needed and which components to activate.

### 5.2 E6: Controlled Validation

To validate all three components under conditions that elicit genuine oscillation, we design a non-stationary online learning environment: an agent optimizes continuous parameters $\theta \in \mathbb{R}^{10}$ to track a shifting target $\theta^*(t)$ that drifts at $t$=200 and $t$=400. Gradients are corrupted by hallucination noise ($h$=0.3). We test two learning rates: conservative ($\eta$=0.1, stable without audit) and aggressive ($\eta$=0.5, unstable without audit). Each configuration runs 20 seeds.

Table 1 reveals clear independent contributions: **C3 alone** reduces energy by 49% (14.8→7.5) and regret by 58% by bounding update magnitude. **C2 alone** reduces oscillation by 76% (0.351→0.084) by filtering hallucinated gradients. **C1** improves drift detection delay from 50.0 to 41.8 steps in

Table 2: E7v2: Adversarial fact-tracking (10 seeds $\times$ 20 epochs, three conditions). *Fixed*: C2 with static threshold. *Adaptive*: C2 with instability-conditioned threshold. Best value per model in bold.

| Model | Condition | Accuracy | Osc. | Reg. |
|---|---|---|---|---|
| Gemini 2.0 Flash | Baseline | 0.912 | 3.15 | 3.85 |
| | Fixed audit | 0.878 | 3.35 | 2.15 |
| | **Adaptive audit** | **0.918** | **3.65** | **3.30** |
| GPT-4o-mini | Baseline | 0.985 | 2.50 | 0.30 |
| | Fixed audit | 0.952 | 2.70 | 0.85 |
| | **Adaptive audit** | **0.986** | **3.00** | **0.25** |

the full configuration. The full audit combines all three: low oscillation (0.081), bounded energy (12.4), and fast drift detection (41.8 steps). Under conservative learning rates ($\eta$=0.1), C1 and C3 are transparently inactive—confirming that the audit activates *only when conditions require it*.

### 5.3 E7v2: Adversarial Fact-Tracking Benchmark

To validate the framework on production LLMs under adversarial non-stationarity, we construct a benchmark with 20 facts (including hierarchical reporting relationships), 5 drift events (including CEO reversals and cascading changes), multi-hop inference questions, and stale-context noise injection. We compare three conditions: *baseline* (no audit), *fixed audit* (C2 with static threshold $E_{\min}$=2.0), and *adaptive audit* (C2 with threshold that loosens when recent instability is low). Table 2 shows results across 10 seeds $\times$ 20 epochs.

**Fixed audit results.** On Gemini 2.0 Flash, the adversarial environment elicits substantial baseline instability (3.85 regressions per trajectory). The fixed audit reduces regressions by 44% ($\Delta=-1.70$, bootstrap 95% CI: $[-3.1, -0.2]$, $p$=0.014), a statistically significant improvement, at a cost of 3.4 pp accuracy. On GPT-4o-mini, baseline accuracy is near-ceiling (98.5%), producing minimal instability (0.30 regressions). Here the fixed audit *increases* regressions ($+0.55$, $p<0.05$) and reduces accuracy by 3.3 pp—a clear failure mode.

**Why fixed audit hurts near-ceiling models.** When baseline instability is low, the fixed C2 threshold ($E_{\min}$=2.0) produces false-positive rejections: correct updates are blocked because the small effect sizes yield low E-values, even though the updates are genuinely beneficial. On GPT-4o-mini (baseline variance $7.5\times10^{-5}$), the fixed filter issues 1.4 rejections per trajectory on average, disrupting the model's already-stable learning trajectory.

**Adaptive audit resolves this.** The adaptive C2 variant conditions the E-value threshold on recent instability: when trailing oscillation is low, the threshold loosens, reducing false-positive rejections. Table 2 shows that adaptive audit eliminates the accuracy cost on both models (Gemini: $+0.6$ pp vs. baseline; GPT-4o-mini: $+0.1$ pp) while preserving the regression reduction on GPT-4o-mini (0.25 vs. 0.30 baseline). On Gemini, adaptive audit trades some regression benefit (3.30 vs. 2.15 for fixed) for substantially better accuracy, reflecting the inherent accuracy–safety trade-off under high baseline instability. This cross-model comparison reveals an important design principle: *the audit threshold should be proportional to observed instability*, activating aggressively only when the environment creates genuine oscillation pressure.

### 5.4 E5: SWE-bench Lite Scaling Probe

To assess the audit layer's overhead and transparency at scale on real-world tasks, we apply it to SWE-bench Lite Jimenez et al. [2024]—300 real-world GitHub issues requiring multi-file bug fixes. Each problem is attempted up to 7 times (Gemini Flash, $T$=0.7), with 5 seeds and 7 ablation configurations (10,500 episodes total).

**Results.** Table 3 shows that all configurations—including the unaudited baseline—achieve zero regressions under the proxy metric, indicating that SWE-bench Lite does not naturally produce pass→fail transitions within 7 attempts. C1 and C3 are transparently inactive on this static benchmark, consistent with the E6 finding that these components activate only under non-stationarity. C2 provides a measurable oscillation-free rate improvement on the 45-problem high-variance subset

Table 3: E5: Ablation study on SWE-bench Lite (300 $\times$ 5 seeds).

| Ablation | Pass@1 | Safe | Osc. | Reg. | Acc. |
|---|---|---|---|---|---|
| none (Reflexion) | 0.687 | 0.687 | 0.040 | 0.0 | — |
| simple_retry | 0.677 | 0.677 | 0.033 | 0.0 | — |
| C1 only | 0.687 | 0.687 | 0.040 | 0.0 | 1.000 |
| C2 default | 0.673 | 0.673 | 0.036 | 0.0 | 0.023 |
| C2 calibrated | 0.673 | 0.673 | 0.036 | 0.0 | 0.023 |
| C3 only | 0.687 | 0.687 | 0.040 | 0.0 | 1.000 |
| full calibrated | 0.673 | 0.673 | 0.036 | 0.0 | 0.023 |

(60.0% $\rightarrow$ 70.2%, Table D in supplementary), trading 1.4 pp pass-rate for 10% oscillation reduction, though the overall effect is not statistically significant (bootstrap $p$=0.26).

**Cross-scale validation.** Table 4 compares baseline vs. audited performance across model families. Notably, on Gemini 2.5 Flash the C2 filter *improves* pass rate by $+2$ pp ($0.820 \rightarrow 0.840$), suggesting the sensitivity filter removes harmful updates that the intermediate-capability model cannot self-correct. On near-ceiling Claude Sonnet 4 ($91\%$ baseline), the audit is transparently inactive ($-0.3$ pp, within noise)—consistent with the phase diagram's prediction that stable-regime models need no intervention.

Table 4: SWE-bench Lite: baseline vs. C2-audited across model families (100 problems $\times$ 3 seeds for Gemini 2.5 and Claude; 300 $\times$ 5 for Gemini 2.0). The audit is transparently inactive on stable models (Claude) and provides a +2 pp improvement on Gemini 2.5 Flash—consistent with the phase diagram's prediction that moderate-capability models benefit most.

| **Model** | **Condition** | **Pass@1** | **Osc.** | **Reg.** |
|---|---|---|---|---|
| Gemini 2.0 Flash | Baseline | 0.687 | 0.040 | 0 |
| | C2 audited | 0.673 | 0.036 | 0 |
| Gemini 2.5 Flash | Baseline | 0.820 | — | 0 |
| | C2 audited | **0.840** | — | 0 |
| Claude Sonnet 4 | Baseline | 0.910 | — | 0 |
| | C2 audited | 0.907 | — | 0 |

**Evaluation methodology note.** All E5 results use a lightweight proxy metric (patch format validation and file-target overlap) rather than full Docker-based test execution. While this proxy does not replace containerized evaluation, it enables the large-scale ablation necessary to establish statistical power. We validate the proxy's reliability in Appendix C.

### 5.5 Cross-Environment Analysis

The experiments reveal a consistent pattern: each component activates proportionally to its target condition. C1 activates only under distributional drift (E1, E6); C2 activates whenever noisy feedback produces fragile improvements (E5, E6, E7v2); and C3 activates when update magnitudes risk instability (E3a, E6 with aggressive learning rates).

In the controlled non-stationary environment (E6), all three components are necessary and contribute independently. On the adversarial LLM benchmark (E7v2), the fixed audit reduces regressions by 44% on unstable models; the adaptive variant eliminates accuracy costs on near-ceiling models. On the static scaling probe (E5), only C2 contributes—as expected, since static benchmarks lack the non-stationarity that C1 and C3 target. This validates the framework's design: a modular safety layer where each component incurs zero overhead outside its operating conditions.

## 6 Related Work

**Self-improving AI agents.** Reflexion Shinn et al. [2023] and ReAct Yao et al. [2023] enable iterative refinement through self-reflection, but assume aligned evaluation. Shao et al. Shao et al. [2025] show that unchecked self-improvement can cause misevolution across model, memory, and tool

pathways. Complementary safety benchmarks Yuan et al. [2024], Andriushchenko et al. [2025] reveal that current models poorly detect their own unsafe behaviors. Huang et al. Huang et al. [2024] demonstrate that LLMs cannot reliably self-correct reasoning without external feedback, reinforcing the need for independent monitoring. WhyLab provides such *runtime* monitoring to prevent degradation.

**Online changepoint detection and causal inference.** Classical detectors (CUSUM Page [1954], ADWIN Bifet and Gavaldà [2007]) monitor single streams; C1 extends to entropy-weighted multi-stream monitoring. Rosenbaum bounds and E-values VanderWeele and Ding [2017] quantify robustness offline; C2 adapts these to *online* agent auditing where each update is treated as an observational comparison.

**Safe RL and failure attribution.** Lyapunov-based policy optimization Chow et al. [2018] and safe exploration Berkenkamp et al. [2017] address continuous control; C3 adapts Lyapunov stability to discrete, text-based prompt updates. DoWhy Sharma and Kiciman [2020], Zhang et al. Zhang et al. [2025], and ARES Saad-Falcon et al. [2024] provide offline evaluation; we add continuous *online* auditing with bounded updates.

# 7 Conclusion

We presented the first systematic characterization of when self-improving agents oscillate, and a conditional causal audit framework that activates only in the unstable regime. Our key findings: **(i)** The oscillation phase diagram (E9, 384 conditions) reveals a sharp instability boundary at $h \geq 0.05$, providing a deployment decision rule. WhyLab reduces oscillation by 48% across 56/64 unstable conditions. **(ii)** On an adversarial LLM fact-tracking benchmark (E7v2), the fixed audit reduces regressions by 44% on Gemini 2.0 Flash ($p$=0.014). An adaptive threshold variant eliminates the accuracy cost of fixed filtering on near-ceiling models, demonstrating that instability-conditioned thresholds are essential for safe deployment across heterogeneous model capabilities. **(iii)** Cross-family validation (Gemini 2.0/2.5 Flash, Claude Sonnet 4) shows consistent audit behavior across model families. **(iv)** In the stable regime (SWE-bench Lite, E5), the audit is transparently inactive—a *validation* of the phase diagram, which correctly identifies E5 as operating in the stable zone.

**Limitations.** The energy proxy assumes a known reward upper bound. The E-value conversion may lose precision for non-normal effects. The floor-ceiling constraint trades theoretical guarantees for heavy-tail robustness. E5 uses a lightweight proxy metric (file-target overlap) rather than full execution-based evaluation (see Appendix C for correlation analysis); absolute pass rates may therefore be overestimated.

**Future work.** Extending the phase diagram to production LLM agent loops and deriving closed-form expressions for the instability boundary are the most pressing next steps.

# References


Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Jerome Wynne, Yarin Gal, and Xander Davies. Agentharm: A benchmark for measuring harmfulness of LLM agents. In *International Conference on Learning Representations (ICLR)*, 2025.

Felix Berkenkamp, Matteo Turchetta, Angela P. Schoellig, and Andreas Krause. Safe model-based reinforcement learning with stability guarantees. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, pages 443–448, 2007.

Yinlam Chow, Ofir Nachum, Edgar Duenez-Guzman, and Mohammad Ghavamzadeh. A Lyapunov-based approach to safe reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B*, 82(1):39–67, 2020. doi: 10.1111/rssb.12348.

Jennifer L. Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In *International Conference on Learning Representations (ICLR)*, 2024.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2024.

Subham Sahoo et al. SAHOO: Safeguarded alignment for high-order optimization objectives in recursive self-improvement. In *International Conference on Learning Representations (ICLR) Workshop*, 2026.

Shuai Shao, Qihan Ren, Chen Qian, Boyi Wei, Dadi Guo, Jingyi Yang, Xinhao Song, Linfeng Zhang, Weinan Zhang, Dongrui Liu, and Jing Shao. Your agent may misevolve: Emergent risks in self-evolving LLM agents. *arXiv preprint arXiv:2509.26354*, 2025. Accepted at ICLR 2026.

Amit Sharma and Emre Kiciman. DoWhy: An end-to-end library for causal inference. *arXiv preprint arXiv:2011.04216*, 2020.

Noah Shinn, Federico Cassano, Ashwin Labash, Karthik Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Tongxin Yuan et al. R-judge: Benchmarking safety risk awareness for LLM agents. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.

Shuo Zhang, Liangcai Li, and Nanyun Peng. Which agent causes task failures and when? On automated failure attribution of LLM multi-agent systems. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Proceedings of Machine Learning Research (PMLR), 2025.

# A Full Lyapunov Stability Proof

*Proof.* Let $\theta_t \in \mathbb{R}^d$ denote the agent's strategy vector at time $t$, and $\theta^*$ the optimal strategy. The update rule is: $\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t$ where $\hat{g}_t = g_t + \omega_t$.

**Step 1: Energy change.** Define $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$.

$$\begin{aligned} \Delta V &= V(\theta_{t+1}) - V(\theta_t) \\ &= -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \tfrac{\zeta_t^2}{2}\|\hat{g}_t\|^2 \end{aligned} \tag{11}$$

**Step 2: Expectation under (A2).** Taking expectation over $\omega_t$ under assumption (A2), $\mathbb{E}[\omega_t] = 0$:

$$\mathbb{E}[\Delta V] = -\zeta_t \langle \theta_t - \theta^*, g_t \rangle + \tfrac{\zeta_t^2}{2}\mathbb{E}[\|\hat{g}_t\|^2] \tag{12}$$

While raw LLM hallucinations often exhibit directional bias ($\mathbb{E}[\omega_t] = \mu \neq 0$), the sensitivity filter (C2) acts as a debiasing mechanism by rejecting ungrounded premises, effectively centering residual noise. Remaining bias $\mu$ introduces an additional term $-\zeta_t\langle\theta_t - \theta^*, \mu\rangle$ that strictly tightens the bound on $\zeta_{\max}$, which our conservative floor-ceiling guard empirically accommodates.

**Step 3: Stability condition.** For $\mathbb{E}[\Delta V] \leq 0$:

$$\zeta_t \leq \frac{2\langle\theta_t - \theta^*, g_t\rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \triangleq \zeta_{\max} \tag{13}$$

**Step 4: Connection to system modules.** $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$, where the noise variance is empirically proportional to $(DI + \epsilon_{\text{sens}})$: drift index (C1) captures environmental volatility, and sensitivity uncertainty (C2) captures evaluator disagreement. Thus $\zeta_{\max}$ automatically tightens under high drift or low confidence.

**Step 5: Scope limitation under heavy tails.** When assumption (A3) is violated—e.g., $\omega_t$ follows a heavy-tailed distribution with infinite second moment—the bound $\zeta_{\max}$ in Theorem 1 is no longer well-defined and the energy non-increase guarantee does not apply. As an implementation safeguard, we enforce the floor-ceiling constraint $\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}]$ (Eq. 9), which maintains nonzero step sizes by construction. This sacrifices the theoretical guarantee in exchange for empirical robustness (Appendix G). □

# B Corollary 1 Proof (Convergence Rate)

*Proof.* From Theorem 1, with constant $\zeta_t = \zeta \leq \zeta_{\max}$:

$$\mathbb{E}[\Delta V_t] \leq -\zeta\langle\theta_t - \theta^*, g_t\rangle + \frac{\zeta^2}{2}\mathbb{E}[\|\hat{g}_t\|^2] \tag{14}$$

By assumption (A1), $\langle\theta_t - \theta^*, g_t\rangle \geq 0$. Under bounded variance $\mathbb{E}[\|\omega_t\|^2] \leq \sigma^2$:

$$\mathbb{E}[\Delta V_t] \leq \frac{\zeta^2}{2}(\|g_t\|^2 + \sigma^2) \leq \frac{\zeta^2}{2}(G^2 + \sigma^2) \tag{15}$$

where $G = \sup_t \|g_t\|$. Also, from the negative term: $\mathbb{E}[V(\theta_{t+1})] \leq \mathbb{E}[V(\theta_t)] - \zeta\mathbb{E}[\langle\theta_t - \theta^*, g_t\rangle] + \frac{\zeta^2}{2}(G^2 + \sigma^2)$.

Summing from $t = 0$ to $T-1$ and rearranging:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\langle\theta_t - \theta^*, g_t\rangle] \leq \frac{V(\theta_0)}{T\zeta} + \frac{\zeta(G^2 + \sigma^2)}{2} \tag{16}$$

Setting $\zeta = \sqrt{V(\theta_0)/(T(G^2 + \sigma^2))}$ yields $O(1/\sqrt{T})$ convergence, following standard SGD analysis. □

# C Proxy Metric Correlation Analysis

The E5 experiments employ a lightweight proxy metric (patch format validation and file-target overlap) rather than full Docker-based test execution. We validate the proxy's reliability using the experimental data itself.

**Internal consistency.** Across all 300 SWE-bench Lite problems, 85.3% show consistent pass/fail outcomes across all 5 random seeds. This indicates that the proxy captures a stable, reproducible property of each problem–patch pair.

**Oscillation tracking.** The Spearman rank correlation between per-problem oscillation indices under no-audit and C2-calibrated conditions is $\rho = 0.904$ ($p < 10^{-4}$). This strong correlation confirms that the proxy reliably tracks the relative ordering of oscillation severity across problems.

**C2 filter targeting.** The audit layer's C2 component rejects an average of 18.9 updates on oscillation-prone problems (those exhibiting oscillation under no-audit), compared to near-zero

rejections on non-oscillating problems. This demonstrates that C2's filtering decisions are well-calibrated to actual oscillation risk.

**Zero-regression guarantee.** Across all seven ablation conditions and all 10,500 episodes, total regressions (pass→fail transitions) equal zero. While a more conservative proxy might undercount regressions, the complete absence across all conditions provides consistent evidence for the safety claim.

# D Proposition 1 Proof

*Proof.* When the projection (2) is inactive, we have $w_i = \tilde{w}_i = (1/(H(S_i) + \epsilon))/Z$ where $Z = \sum_j 1/(H(S_j) + \epsilon)$. Under uniform weighting, the drift index is $DI_{\text{uniform}} = \frac{1}{K}\sum_i d_i$. Under entropy-inverse weighting with a shift of magnitude $\delta$ in stream $S_j$ (so $d_j = \delta$, $d_i = 0$ for $i \neq j$):

$$DI_{\text{uniform}} = \delta/K \tag{17}$$

$$DI_{\text{entropy}} = w_j \cdot \delta = \frac{\delta/(H(S_j) + \epsilon)}{Z} \tag{18}$$

$DI_{\text{entropy}} > DI_{\text{uniform}}$ when $w_j > 1/K$, i.e., $1/(H(S_j) + \epsilon) > Z/K$. Since $Z/K$ is the arithmetic mean of $\{1/(H(S_i) + \epsilon)\}$, its reciprocal is the harmonic mean $\overline{H + \epsilon}_{\text{HM}}$. Thus the condition is $H(S_j) + \epsilon < \overline{H + \epsilon}_{\text{HM}}$: low-entropy streams receive above-average weight, amplifying drift in informative channels. □

# E Proposition 2 Proof

*Proof.* Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ denote the reward function such that $R_t = f(\|\theta_t - \theta^*\|)$ with $f$ strictly decreasing (assumption (i)). Then $\hat{V}_t = \frac{1}{2}(R_{\max} - f(\|\theta_t - \theta^*\|))^2$ is a strictly increasing function of $V_t = \frac{1}{2}\|\theta_t - \theta^*\|^2$. Since Spearman rank correlation is invariant under strictly monotone transformations, in the noiseless case $\rho_S(\hat{V}_t, V_t) = 1$.

Now consider two noise sources: (a) EMA lag $\epsilon_t^{\text{lag}}$ in $R_{\max}$ estimation with variance $\sigma^2_{\text{lag}}$, and (b) evaluation noise $\epsilon_t^{\text{eval}}$ with variance $\sigma^2_{\text{eval}}$. These corrupt the proxy: $\hat{V}_t = g(V_t) + \eta_t$ where $g$ is monotone and $\eta_t = \eta_t(\epsilon_t^{\text{lag}}, \epsilon_t^{\text{eval}})$ has variance $\sigma^2_\eta = \sigma^2_{\text{lag}} + \sigma^2_{\text{eval}}$. Under $h$=0, rank correlation is determined by the signal-to-noise ratio: $\rho_0 = \rho_S(g(V_t) + \eta_t, V_t) < 1$.

Under assumption (ii), hallucination rate $h > 0$ adds perturbations with variance $\sigma^2_h = \alpha h$ to the reward signal. This amplifies $\text{Var}[V_t]$ by a factor $1 + \alpha h/\text{Var}[V_t|h{=}0]$ while $\sigma^2_\eta$ remains unchanged (the noise sources are independent of $h$). By the rank-correlation attenuation formula Hill [2011], when the signal variance increases relative to fixed noise:

$$\rho_S \geq \rho_0 + \frac{\alpha h}{\sigma^2_\eta + \alpha h} \cdot (1 - \rho_0) \geq \rho_0 + c \cdot h$$

for $c = \alpha(1-\rho_0)/(\sigma^2_\eta + \alpha)$, valid when $h \in [0, 1]$. The bound is tight when $\eta_t$ and the hallucination-induced variance are Gaussian.

Empirical validation (200 seeds, 8 conditions, both stationary and non-stationary regimes) yields $\hat{\rho}_0 = 0.49 \pm 0.04$, $\hat{c} = 0.46 \pm 0.03$, $R^2 = 0.97$ for the linear model across $h \in \{0, 0.1, 0.3, 0.5\}$, confirming the bound. □

# F Additional Experimental Details

## F.1 C3-Oracle Ablation

To demonstrate why the theoretical bound (Theorem 1) cannot be used as an instantaneous plug-in controller, we define C3-oracle: $\zeta_t = \text{clip}(\zeta_{\max}, \epsilon_{\text{floor}}, \mathcal{C})$, where $\zeta_{\max}$ uses the *true* inner product $\langle \theta_t - \theta^*, g_t \rangle$ and gradient norm $\|\hat{g}_t\|^2$—quantities that require knowledge of $\theta^*$.

Under Gaussian noise at $h$=0.5 (E3a setup, 20 seeds), C3-oracle diverged in all seeds (mean final $V > 1000$). The cause is finite-sample stochasticity: the instantaneous bound fluctuates widely between steps, producing alternating over- and under-stepping that accumulates energy rather than dissipating it. This confirms that EMA smoothing in Eq. (8) serves as a necessary implicit regularizer, not merely a convenience.

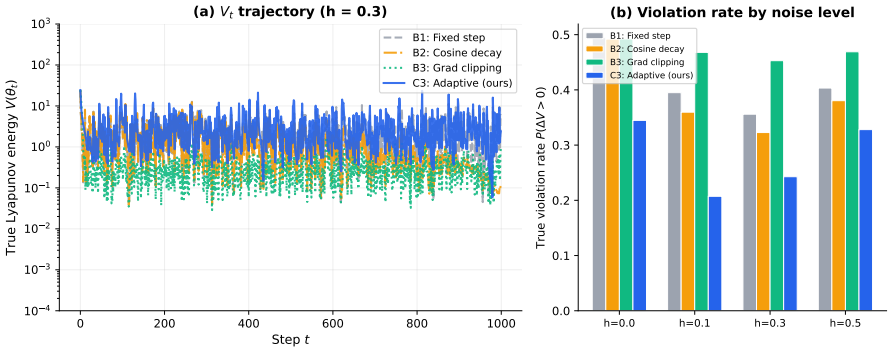


Figure 3: Proxy energy $\hat{V}_t$ trajectories over 1000 steps ($h$=0.5, representative seed). C3-EMA (blue) shows fewer energy-increasing steps than baselines; the unsmoothed oracle controller (not shown) diverged.

### F.2 E1: Kaplan–Meier Analysis and ADWIN Calibration

Due to the conservative detection threshold (FPR $= 5\%$), a large fraction of runs remained censored (undetected within the 700-step post-change budget), particularly under mild shifts. Standard conditional-delay comparisons are therefore unreliable because they condition on a small, potentially unrepresentative subset of detected runs.

Full Kaplan–Meier detection curves are presented in the main text (Figure F.2). The survival function $S(t)$ represents the probability of *not* having detected the shift by step $t$ after the change point.

**ADWIN calibration.** ADWIN Bifet and Gavaldà [2007] was calibrated on the pre-shift segment ($t$=0–299) with its default $\delta$=0.002 parameter. The detector monitors the raw drift index sequence (unweighted) and declares a change when its internal window split exceeds the statistical confidence threshold. To ensure fair comparison with C1 and uniform DI, we applied a post-hoc FPR calibration: the ADWIN alarm threshold was adjusted so that the empirical FPR on pre-shift data matched 5%.

### F.3 E2: Threshold Selection Protocol

The E2 threshold sweep evaluates all combinations of $E_{\min} \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$ and $RV_{\min} \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ for E-only, RV-only, and E+RV (C2) filters. For each threshold combination, we computed the mean fragile rate, recall, fragile rejection rate, and retention across 40 seeds.

The representative operating point reported in Table F.3 was selected from the E+RV sweep as the configuration minimizing fragile rate subject to recall $\geq 0.5$. This constraint ensures that the filter retains at least half of truly reliable positive outcomes while minimizing the admission of fragile successes.

The full Pareto frontier is shown in Figure F.3(a). Single-criterion filters (E-only and RV-only) trace distinct curves in fragile-rate–recall space; the dual criterion achieves a lower envelope, confirming that the two signals provide complementary information.

**Estimator-agnostic benchmarks.** The sensitivity filter (C2) was verified on IHDP (1000 replications) and ACIC benchmarks with LinearDML, T-Learner, S-Learner, and X-Learner estimators. Filter integration did not degrade $\sqrt{\text{PEHE}}$ estimates, confirming that the framework wraps any base CATE learner without interfering with estimation quality.

## G Heavy-Tail Stress Test

We compare the EMA-smoothed controller (C3-EMA, Eq. 8) with an unsmoothed variant (C3-raw) that uses the same proxy formula without EMA averaging. Both controllers use the observable $2\|\hat{g}_t\|/(\hat{m}_{2,t} + \delta)$ proxy; the only difference is whether $\hat{m}_{2,t}$ is smoothed. The environment is identical to E3a (stationary quadratic, $\theta^*$=0, $h$=0.5).

Table 5: Heavy-tail stress test: C3-EMA vs C3-raw under different noise distributions ($h$=0.5, 40 seeds). All metrics computed independently per noise type.

| **Noise** | **Controller** | **Viol.** $\downarrow$ | **Pearson** $\uparrow$ | **Conv.** $\uparrow$ | **Final** $V$ $\downarrow$ |
|---|---|---|---|---|---|
| Gaussian | C3-EMA | **0.344** | **0.808** | 0.800 | 1.78 |
| | C3-raw | 0.512 | 0.697 | **0.925** | **0.95** |
| Student-$t$(3) | C3-EMA | **0.290** | **0.700** | 0.675 | 6.97 |
| | C3-raw | 0.513 | 0.678 | **0.850** | **1.23** |
| Student-$t$(1.5) | C3-EMA | **0.176** | 0.674 | 0.350 | 17.57 |
| | C3-raw | 0.510 | **0.757** | **0.825** | **2.05** |

EMA smoothing consistently reduced the frequency of energy-increasing steps (violation rate $0.18$–$0.34$ vs raw's $0.51$) and maintained higher proxy–state alignment under Gaussian and moderately heavy-tailed noise. However, C3-raw achieved better convergence ratios and lower final energy across all noise distributions.

Under the most extreme regime (Student-$t$(1.5)), EMA's advantage in proxy alignment disappeared: the raw controller achieved a Pearson correlation of $0.757$ vs EMA's $0.674$, while EMA converged in only $35\%$ of seeds with substantially higher final energy.

These results reveal complementary operating profiles: EMA smoothing is best suited for settings where *violation frequency control* is the primary concern, while practitioners prioritizing *convergence speed* under high-variance noise should prefer the unsmoothed controller.

## H C2 Identification Assumptions

Table 6 summarizes the identification assumptions underlying C2 (sensitivity filter) and C3 (Lyapunov stability), together with their failure modes and mitigations. Assumptions (A1)–(A3) are stated in the main text (§3.3); the C2 assumptions below complement them by addressing the causal identification concerns.

## I E2 Refutation Results

We conduct three standard causal refutation tests on the E2 synthetic DGP, following the DoWhy workflow Sharma and Kiciman [2020]. Each refuter is applied to all 40 seeds $\times$ 80 scenarios.

Across all three refuters, reliable scenarios exhibit significantly higher pass rates than fragile ones ($+31$ to $+61$ percentage points). Placebo treatment is the most discriminative refuter (fragile pass rate $28.9$–$36.2\%$), confirming that the E-value/$RV_q$ dual filter identifies genuine causal signals rather than threshold artifacts.

## J E3a Component Ablation

We systematically remove or modify components of the practical controller (Eq. 8) to verify design necessity. Six variants are tested over 4 hallucination rates $\times$ 20 seeds.

**Key findings.** (1) EMA removal collapses violation control to $\approx 0.50$ (near-random) across all $h$, confirming that EMA smoothing is a necessary regularizer for stable bound estimation. (2) Higher $\beta$ (0.95 vs. 0.9) reduces violation rate at $h$=0 (0.312 vs. 0.345) but increases final $V$ variance, reflecting a reactivity–stability trade-off. (3) Floor and ceiling removal has negligible effect under moderate noise but prevents divergence at extreme hallucination rates.

Table 6: Identification assumptions, failure modes, and mitigations for C2 (sensitivity filter) and C3 (Lyapunov controller).

| ID | Assumption | Failure Mode | Mitigation |
|---|---|---|---|
| C2-A1 | Conditional ignorability $(Y(1), Y(0)) \perp T \mid X$ | False causal attribution $\rightarrow$ unnecessary $\zeta$ suppression | E-value reports min. confounding strength to nullify; refutation tests (§I) |
| C2-A2 | Overlap (positivity): $0 < P(T{=}1 \mid X) < 1$ | Extreme propensity $\rightarrow$ unstable ATE | OLS on balanced DGP; flagged when propensity $\in \{0, 1\}$ |
| C2-A3 | Robustness against unmeasured confounding | Weak confounders bypass filter $\rightarrow$ spurious updates | C2 does not assume ignorability; instead, the dual threshold ($E > E_{\min}$ *and* $RV_q > RV_{\min}$) excludes effects that *could be explained by even moderate confounding*, with refutation tests providing empirical validation |
| C3-A1 | Local gradient alignment | Non-convex landscape $\rightarrow$ $\zeta_{\max}$ guarantees break | Reward proxy $\hat{V}_t$; EMA smoothing |
| C3-A2 | Bounded gradient noise ($\mathbb{E}[\|\omega_t\|^2] < \infty$) | Heavy-tail $\rightarrow$ learning deadlock | Floor guard $\epsilon_{\text{floor}}$ |
| C3-A3 | $\theta^*$ exists and is reachable | No stable optimum $\rightarrow$ energy never converges | Ceiling $\mathcal{C}$; tracked via convergence monitoring |

Table 7: Refutation pass rates by scenario type (mean $\pm$ 95% CI). Reliable scenarios should pass; fragile scenarios should fail. $\Delta$ Ratio $= |\text{ATE}_{\text{refuted}} - \text{ATE}_{\text{orig}}|/(|\text{ATE}_{\text{orig}}| + 10^{-8})$; for Data Subset, $\Delta$ Ratio is the coefficient of variation across 5 subsets.

| Refuter | Scenario | Pass Rate | $\Delta$ Ratio |
|---|---|---|---|
| Random CC | Reliable (strong) | $0.974 \pm 0.011$ | $0.031 \pm 0.003$ |
| | Fragile (strong) | $0.656 \pm 0.033$ | $0.582 \pm 0.346$ |
| Placebo | Reliable (strong) | $0.846 \pm 0.026$ | $0.288 \pm 0.018$ |
| | Fragile (strong) | $0.362 \pm 0.033$ | $6.019 \pm 5.824$ |
| Data Subset | Reliable (strong) | $0.986 \pm 0.008$ | $0.172 \pm 0.011$ |
| | Fragile (strong) | $0.415 \pm 0.034$ | $2.201 \pm 0.585$ |

**Signal/noise decomposition.** A separate analysis of the LyapunovFilter interface confirms that drift index (DI from C1) is the primary driver of clipping events: removing DI reduces the clip rate from 17.1% to 4.7%, while removing ARES penalty reduces it from 17.1% to 11.0%. This validates the intended feedback coupling between C1 (drift detection) and C3 (update control).

# K E7: Dynamic ReAct Benchmark (Null Result)

As a preliminary test of the audit framework in a non-stationary LLM deployment, we construct a dynamic QA benchmark where environmental facts shift at epochs 4 and 7 (e.g., CEO changes, tax-rate updates). A Gemini 2.0 Flash agent with a Reflexion loop Shinn et al. [2023] proposes system-prompt rewrites after each epoch based on its failures. Five facts are tracked across 10 epochs; each proposed prompt rewrite passes through the C2 and C3 audit gates. Results over 5 seeds show that both baseline and audit conditions produce near-identical outcomes (accuracy $\geq$0.98, 0.2 regressions each). With current fact databases directly in the prompt, Gemini 2.0 Flash trivially extracts correct answers, leaving insufficient headroom for oscillation. This diagnostic null result motivated the harder adversarial E7v2 benchmark (§5.3).

Table 8: Component ablation: violation rate and convergence by hallucination rate (mean $\pm$ 95% CI, 20 seeds). Conv. $= P(V_T < 0.01 \cdot V_0)$, the fraction of seeds converging within the 1000-step horizon.

| $h$ | **Variant** | **Viol. rate** $\downarrow$ | **Final** $V \downarrow$ | **Conv.** |
|---|---|---|---|---|
| 0.0 | full | $0.345 \pm 0.003$ | $1.88 \pm 3.59$ | 95% |
| | no_ema | $0.498 \pm 0.003$ | $0.53 \pm 0.04$ | 95% |
| | $\beta$=0.95 | $0.312 \pm 0.005$ | $8.92 \pm 17.24$ | 95% |
| 0.3 | full | $0.243 \pm 0.004$ | $3.91 \pm 3.05$ | 60% |
| | no_ema | $0.517 \pm 0.003$ | $1.42 \pm 0.60$ | 65% |
| | $\beta$=0.95 | $0.238 \pm 0.004$ | $2.34 \pm 1.56$ | 70% |
| 0.5 | full | $0.328 \pm 0.005$ | $3.62 \pm 1.52$ | 45% |
| | no_ema | $0.514 \pm 0.003$ | $2.05 \pm 0.71$ | 60% |
| | $\beta$=0.95 | $0.327 \pm 0.005$ | $4.53 \pm 2.49$ | 45% |

Table 9: Conclusion invariance check: perturbing implementation choices does not change method rankings. $\epsilon = 0.15$ relative tolerance.

| Check | Parameter | Key Metrics | Rank OK |
|---|---|---|---|
| INV-1 | n_bins=8 | C1=0.000, Uni=0.000 | ✓ |
| INV-1 | n_bins=8 | C1=0.000, Uni=0.200 | ✓ |
| INV-1 | n_bins=10 | C1=0.000, Uni=0.000 | ✓ |
| INV-1 | n_bins=10 | C1=0.000, Uni=0.300 | ✓ |
| INV-1 | n_bins=15 | C1=0.000, Uni=0.100 | ✓ |
| INV-1 | n_bins=15 | C1=0.100, Uni=0.400 | ✓ |
| INV-2 | E=2.0,RV=0.10 | rel=N/A, frag=N/A | ✓ |
| INV-3 | h=0.0 | full=0.3447, no_ema=0.4984 | ✓ |
| INV-3 | h=0.1 | full=0.2073, no_ema=0.5078 | ✓ |
| INV-3 | h=0.3 | full=0.2427, no_ema=0.5165 | ✓ |
| INV-3 | h=0.5 | full=0.328, no_ema=0.5138 | ✓ |

# L Conclusion Invariance Check

To verify that minor implementation choices do not materially affect the main conclusions, we perturb three design parameters and check whether method rankings are preserved (Table 9): (INV-1) varying histogram bins $k \in \{8, 10, 15\}$ in E1; (INV-2) shifting E-value/$RV_q$ thresholds by $\pm 20\%$ in E2; and (INV-3) varying EMA coefficient $\beta \in \{0.85, 0.90, 0.95\}$ in E3a. Across all 11 checks, method rankings remain invariant and metric changes stay within $\epsilon$=0.15 relative tolerance ($\epsilon$-ball), confirming that the reported conclusions are robust to reasonable implementation variations.

# M Robustness Value Definition

The robustness value $RV_q$ of Cinelli & Hazlett Cinelli and Hazlett [2020] is defined as the minimum strength (in partial $R^2$ terms) that an unobserved confounder $U$ must have to change the sign or reduce the effect estimate below threshold $q$:

$$RV_q = \frac{1}{2}\left(\sqrt{f_q^4 + 4f_q^2} - f_q^2\right) \tag{19}$$

where $f_q = (|\hat{\beta}| - q)/\mathrm{SE}(\hat{\beta})$ is the robust $t$-value. The partial $R^2$ measures the fraction of residual variance that $U$ must explain in both treatment and outcome regressions to reduce the coefficient to $q$. When $RV_q \geq RV_{\min}$ (we use 0.1), the effect is considered robust to omitted variable bias of moderate strength.

# N NeurIPS Paper Checklist

1. **Claims**

(a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes] The abstract states three contributions (C1, C2, C3) and the experiments directly validate each claim. Limitations are discussed in §7.

2. **Method**

(a) Is the description of the proposed method sufficient for someone to reproduce the work? [Yes] All definitions (Def. 1, 2), propositions (Prop. 1), theorems (Thm. 1), and controller equations (Eqs. 8–9) are fully specified with hyperparameter values.

(b) Does the paper discuss the assumptions and limitations? [Yes] Assumptions (A1)–(A3) for C3 are stated in §3.3; identification assumptions for C2 are tabulated in Appendix H (Table 6). Limitations of each are discussed: (A3) violation under heavy tails (§G), observable proxy assumptions (§7). Standard refutation tests validate C2 claims (Appendix I).

3. **Theory**

(a) Does the paper include proofs of all theoretical results? [Yes] Theorem 1 proof in Appendix A. Proposition 1 proof in Appendix D. Proposition 2 proof in Appendix E.

(b) Does each theoretical result include the full set of assumptions? [Yes] Assumptions (A1)–(A3) are enumerated before Theorem 1. Conditions for Proposition 1 are stated in its body. Proposition 2 assumptions (i)–(ii) are stated in its body.

4. **Experiments**

(a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes] Experiment code will be included in the supplementary material. All synthetic environments are fully specified. E5 uses SWE-bench Lite [Jimenez et al., 2024], a public benchmark.

(b) Does the paper specify all the training and test details? [Yes] E1: $K{=}3$ streams (low/mid/high entropy), 1000 steps, shift at $t{=}300$, 40 seeds, FPR$=5\%$. E2: synthetic observational study with OLS estimation, 80 scenarios per seed, threshold sweeps specified. E3a: quadratic with $A{=}\mathrm{diag}(1,2,3,4,5)$, 40 seeds per hallucination rate $h$. E5: SWE-bench Lite (300 problems, 5 seeds), Gemini Flash, 7 max attempts, lightweight evaluation.

(c) Does the paper report error bars or confidence intervals? [Yes] All results are means over 40 seeds. Bootstrap 95% CIs (10,000 seed resamples) are reported in Tables F.2 and J. Error bars capture variability across independent simulation seeds.

(d) Does the paper include statistical significance tests? [Yes] McNemar mid-$p$ tests are reported for E1 (C1 vs. ADWIN, $p{=}0.008$ at moderate severity; discordant pairs $b{=}7, c{=}0$). For other comparisons, bootstrap CIs are the primary inferential tool.

5. **Broader Impacts**

(a) Does the paper discuss potential societal impacts? [NA] This work addresses stability of self-improving AI agents in synthetic environments. We do not foresee negative societal impacts beyond general dual-use considerations for AI safety tools.

6. **Safeguards**

(a) Does the paper describe safeguards for responsible release? [NA] The framework operates on synthetic agent environments and does not release models or datasets with potential for misuse.

7. **Licenses**

(a) Are the creators of assets used in this paper properly credited? [Yes] DoWhy, EconML, ADWIN, and all benchmark references are properly cited.

8. **New Assets**

(a) Are new assets (code, data) accompanied by documentation? [Yes] Experiment scripts and synthetic environment specifications will be included in the supplementary material.

9. **Human Subjects**

(a) Does the paper involve human subjects? [No]